

Marketing Funnel & Conversion Performance Analysis

1. Introduction

This report presents a detailed analysis of an e-commerce marketing funnel using real-world user behavior data. The objective of the analysis is to understand how users progress through key funnel stages — **Visitors → Leads → Customers** — and to identify conversion drop-offs, performance gaps, and opportunities to improve lead-to-customer conversion.

The analysis is aligned with common practices used in startups, SaaS companies, and digital marketing teams, where optimizing funnel performance directly impacts revenue growth.

2. Dataset Overview

- **Dataset Used:** E-commerce User Behavior Dataset (Multi-category Store)
- **Source:** Kaggle
- **Time Period Analyzed:** November 2019
- **File Used:** 2019-Nov.csv

Key Columns Used:

- `user_id` – Unique identifier for users
- `event_type` – User action (view, cart, purchase)
- `price` – Product purchase price (used for revenue analysis)

The dataset contains millions of event-level records representing real user interactions on an e-commerce platform.

3. Data Cleaning & Preparation

The following preprocessing steps were applied:

1. Removed records with missing `user_id` values to ensure accurate user-level analysis
2. Filtered relevant funnel events: view, cart, and purchase
3. Aggregated data at the **unique user level** for funnel calculations

These steps ensured clean, structured data suitable for funnel and conversion analysis.

4. Funnel Definition

The marketing funnel was defined as:

1. **Visitors:** Users who viewed at least one product (view event)
2. **Leads:** Users who added a product to cart (cart event)
3. **Customers:** Users who completed a purchase (purchase event)

Each user was counted once per stage to avoid duplication.

5. Funnel Metrics & Conversion Rates

Funnel Stage Counts:

- **Visitors:** Users who viewed products
- **Leads:** Users who added products to cart
- **Customers:** Users who completed purchases

Conversion Rates Calculated:

- **Visitor → Lead Conversion Rate**
Percentage of visitors who added items to cart
- **Lead → Customer Conversion Rate**
Percentage of cart users who completed purchases
- **Overall Conversion Rate**
Percentage of visitors who became customers

These metrics provide a clear view of funnel efficiency at each stage.

6. Funnel Visualization

A bar chart was created using Matplotlib to visually represent the funnel stages and user counts. This visualization makes it easy to identify where the largest user drop-offs occur and supports stakeholder-friendly interpretation.

7. Drop-Off Analysis

Key Drop-Off Points Identified:

1. **Visitor → Lead Drop-Off**
A significant portion of users viewed products but did not add them to the cart.

2. Lead → Customer Drop-Off

A noticeable number of users abandoned the funnel after adding products to the cart.

Interpretation:

- The highest drop-off typically occurs at the **top and middle of the funnel**, which is common in e-commerce platforms.
 - These drop-offs represent major opportunities for optimization.
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8. Revenue Insights

Additional purchase-level analysis was conducted:

- **Total Revenue:** Calculated using purchase prices
- **Average Order Value (AOV):** Mean value of completed purchases

These metrics help connect funnel performance directly to business outcomes.

9. Key Business Insights

- High product views do not automatically translate into cart actions, indicating potential UX or pricing friction
 - Cart abandonment remains a critical issue affecting final conversions
 - Even small improvements in mid-funnel conversion rates can significantly increase revenue
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10. Actionable Recommendations

To Improve Visitor → Lead Conversion:

- Enhance product page design (clear CTAs, high-quality images, reviews)
- Improve site speed and mobile responsiveness
- Offer limited-time discounts or free shipping indicators

To Improve Lead → Customer Conversion:

- Implement cart abandonment emails or notifications
- Simplify checkout process and reduce form friction

- Provide multiple secure payment options

Overall Funnel Optimization:

- Retarget high-intent users using ads or email campaigns
- Analyze product categories with highest drop-offs for targeted improvements
- A/B test pricing, promotions, and checkout layouts

11. Conclusion

This funnel analysis successfully identified key conversion metrics, drop-off points, and improvement opportunities within an e-commerce marketing funnel. The insights generated from this analysis can be directly used by marketing and growth teams to optimize user journeys, improve conversions, and increase revenue.

The project demonstrates strong skills in **funnel analysis, conversion metrics, data-driven insights, and business-focused recommendations**, fulfilling all Task-3 deliverable requirements.
